

Applied Generative AI

Session 1: The Story of Generative AI – From Rules to Intelligence

Learning Objectives

By the end of this session, you will be able to:

- Understand why Generative AI is considered a technological revolution.
- Differentiate between Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI.
- Explain why Transformers changed Natural Language Processing.
- Understand the concept of Foundation Models.
- Describe how modern AI applications are built.

Introduction

Technology has continuously evolved to solve increasingly complex problems. The Internet connected people across the globe, smartphones made computing portable, and cloud computing made powerful infrastructure accessible to everyone.

Today, Generative AI represents the next major technological shift.

Unlike traditional software that follows predefined instructions, Generative AI can create new content such as text, images, code, music, and videos. It assists users in solving problems, writing software, answering questions, summarizing documents, and automating knowledge work.

Many experts compare the rise of Generative AI to the emergence of the Internet because it is expected to transform almost every industry.

As developers, our goal is not merely to use AI tools like ChatGPT or Gemini. Our objective is to understand how they work and build intelligent applications using these technologies.

Evolution of Computing

Computing has evolved through several major revolutions.

Era	Innovation
Traditional Computing	Rule-based software

Internet	Connected computers worldwide
Mobile Computing	Computing anywhere
Cloud Computing	On-demand infrastructure
Artificial Intelligence	Machines learn patterns
Generative AI	Machines generate content

Each revolution expanded what computers could accomplish. Generative AI goes one step further by enabling computers to assist with creative and knowledge-based tasks.

Traditional Programming

For decades, software was built using explicit instructions written by programmers.

The relationship can be represented as:

Rules + Data = Output

For example, consider a simple program to identify a cat.

IF animal has

4 legs

AND whiskers

AND tail

THEN

Cat

This approach works well for simple problems.

However, real-world situations are rarely so straightforward.

What if the image is blurry?

What if the cat has no tail?

What if the lighting is poor?

Soon, the number of rules becomes extremely difficult to manage.

This limitation motivated researchers to explore a different approach.

Artificial Intelligence

Artificial Intelligence (AI) is the branch of computer science focused on creating systems capable of performing tasks that normally require human intelligence.

Examples include:

- Speech Recognition
- Face Detection
- Language Translation
- Medical Diagnosis
- Autonomous Vehicles
- Recommendation Systems

AI is a broad field that contains many specialized areas, including Machine Learning and Deep Learning.

Machine Learning

Instead of writing rules manually, Machine Learning allows computers to learn patterns from examples.

Rather than programming every possible situation, we provide historical data and expected outputs.

For example:

Image	Label
Dog Image	Dog

Cat Image	Cat
Bird Image	Bird

After seeing thousands or even millions of examples, the model learns how to recognize new images on its own.

The relationship now becomes:

Data + Correct Answers = Learning Algorithm → Model

This was a major breakthrough because the computer discovered patterns automatically instead of relying on manually written rules.

Deep Learning

Traditional Machine Learning often required engineers to manually define important features.

For example, when recognizing dogs, engineers might specify:

- Ear shape
- Tail length
- Fur color
- Body size

Deep Learning eliminated much of this manual effort.

Neural Networks automatically learn useful features directly from data.

This significantly improved performance in tasks such as:

- Image Recognition
- Speech Recognition
- Language Translation
- Medical Imaging

Deep Learning became the foundation for modern Generative AI.

The Language Problem

Although Deep Learning achieved remarkable success in image processing, language remained a difficult challenge.

Consider these two sentences:

I deposited money in the bank.

We sat beside the bank of the river.

The word **bank** has different meanings depending on context.

Humans understand this naturally.

Earlier AI systems struggled because they processed words with limited understanding of surrounding context.

Researchers needed a better solution.

Transformers and Attention

In 2017, researchers introduced the Transformer architecture through the paper "**Attention Is All You Need.**"

The key innovation was **Attention**.

Attention allows the model to determine which words are most important when interpreting a sentence.

For example:

The trophy doesn't fit into the suitcase because it is too big.

Humans immediately understand that **it** refers to the trophy.

Transformers learned to make similar contextual connections.

This innovation dramatically improved language understanding and became the foundation for modern Large Language Models.

GPT and Foundation Models

GPT (Generative Pre-trained Transformer) is based on the Transformer architecture.

Its primary task is surprisingly simple:

Predict the next token.

Suppose the input is:

The capital of France is

The model predicts:

Paris

After generating one token, it predicts the next one, and then the next, gradually constructing an entire response.

Although the underlying process is simple, training on trillions of words enables GPT to generate high-quality text, write software, summarize information, and answer questions.

Models such as GPT, Gemini, Claude, and Llama are called **Foundation Models** because they can perform many different tasks using the same underlying model.

Modern AI Applications

Most production AI systems involve much more than an LLM.

A typical application looks like this:

User

|

Frontend

|

Backend

|

Knowledge Base

|

Retriever

|

Large Language Model

|

Tools / APIs

|

Response

Developers build these complete systems rather than simply calling an AI model.

Open Models vs Closed Models

Closed Models	Open Models
GPT	Llama
Claude	Mistral
Gemini	Gemma

Closed Models

- Managed by vendors
- Easy to use
- Excellent performance
- API-based access

Open Models

- Can be self-hosted
- Customizable
- Better privacy
- Require more engineering effort

The Future: Small Language Models

A common misconception is that larger models are always better.

In reality, many business applications require only small, efficient models.

Examples include:

- Microsoft Phi
- Google Gemma
- IBM Granite
- Small Llama models

These models are:

- Faster
- Less expensive
- Easier to deploy locally
- Suitable for enterprise applications

Key Takeaways

By the end of this session, you should remember the following:

1. Traditional software follows rules; AI learns patterns.
2. Machine Learning learns from data instead of manually written rules.
3. Deep Learning automatically discovers useful features.
4. Transformers introduced Attention, dramatically improving language understanding.
5. GPT generates text by predicting one token at a time.
6. Foundation Models can perform many tasks using the same model.
7. Modern AI applications combine LLMs with software engineering, retrieval systems, databases, and external tools.

Glossary

Term	Definition
Artificial Intelligence	Systems that perform tasks requiring human intelligence.
Machine Learning	A subset of AI where models learn from data.
Deep Learning	Neural networks that automatically learn complex patterns.
Transformer	Neural architecture based on the Attention mechanism.

Attention	Mechanism that helps models focus on relevant information.
LLM	Large Language Model trained on massive text datasets.
Foundation Model	A general-purpose AI model that supports multiple downstream tasks.
Token	The smallest unit of text processed by an LLM.

What's Next?

In **Session 2**, we will answer one of the most common questions about AI:

How does a Large Language Model actually work?

You will learn about **tokens, embeddings, context windows, attention, training, inference, and the complete pipeline that transforms human language into AI-generated responses.**